

Net ecosystem

This is just the beginning of something big.

chronology



Only runs on windows
platform

mostly for cross
platform e.g mac,linux
,decoupling from
windows

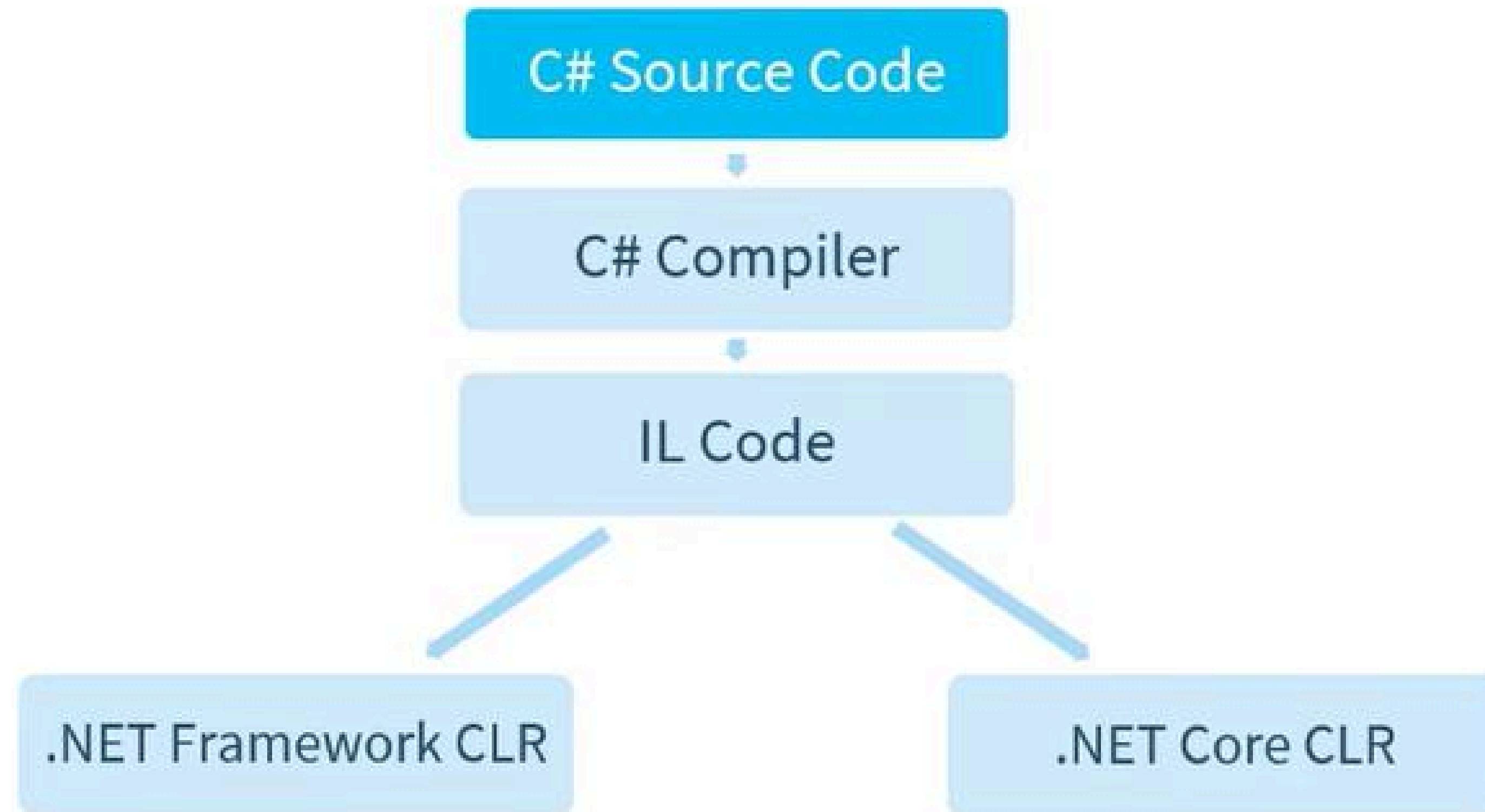
.Net ecosystem : what this means

- **Languages** : c# , f#
- **Runtime** - core clr
- **Libraries** : BCL , 3rd party

When we write a C# program, it gets compiled into Intermediate Language (IL) code, which is platform-independent.

The CLR then uses a Just-In-Time (JIT) compiler to convert the IL code into machine-specific code while the program runs.

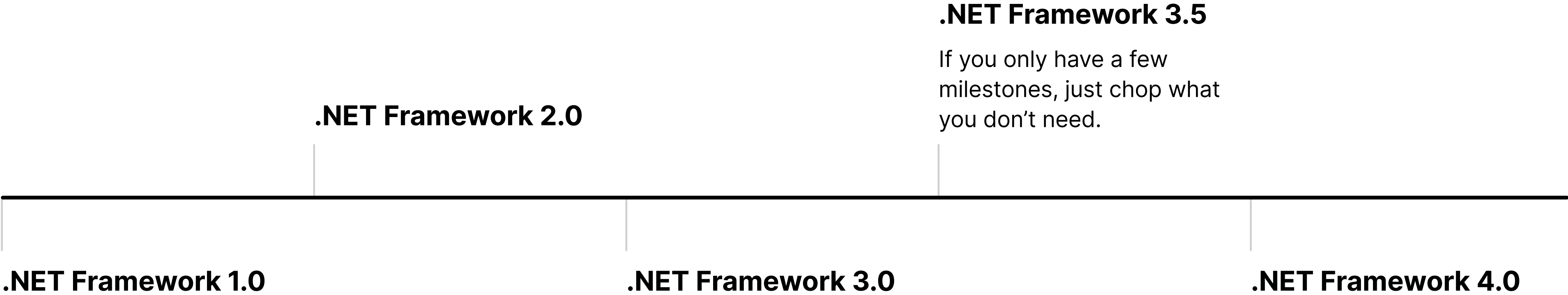
THE COMMON LANGUAGE RUNTIME (CLR)



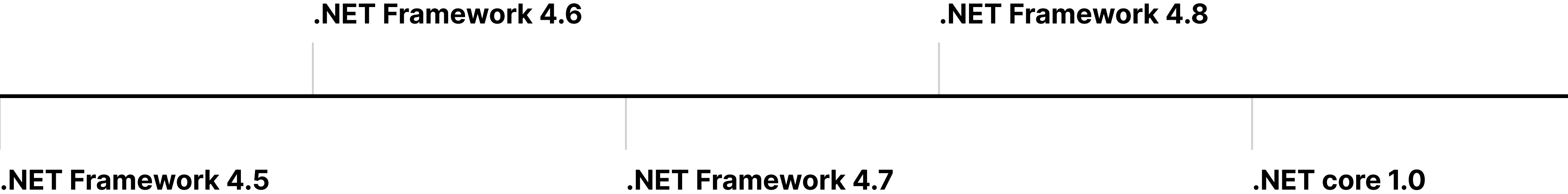
The .NET Framework 1.0 introduced the .NET CLR, the .NET base class libraries, ASP.NET, and, most importantly, the first version of Visual Studio .NET.

Asp.NET is Web developement framework made by micrososft

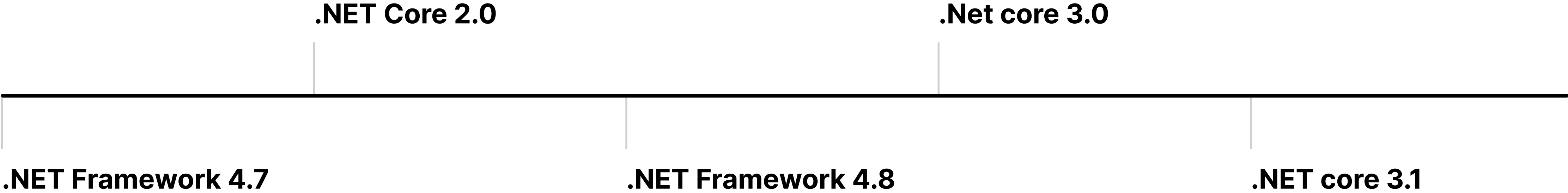
Net Timeline



Net Timeline



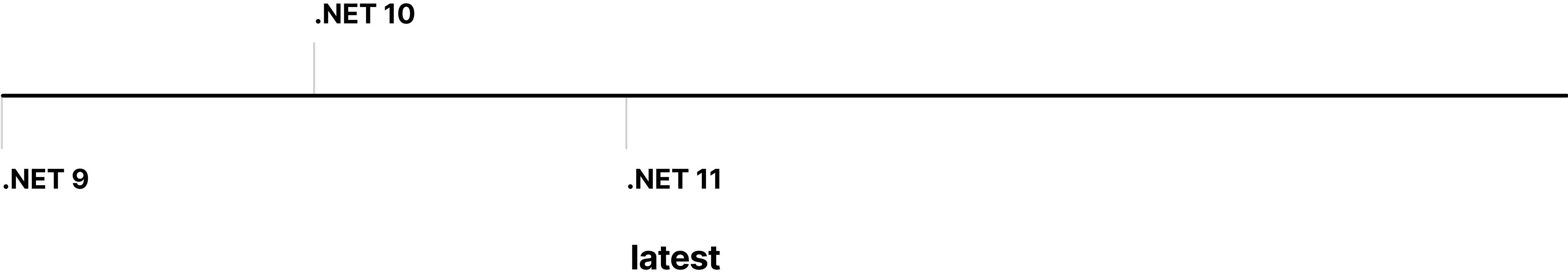
Net Timeline



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.NET Standard

The .NET Standard is a formal specification that different frameworks implement. It describes the API surface a framework has to implement to be compatible with a .NET Standard.

The idea behind .NET Standard was to allow code sharing between projects targeting different frameworks, such as

- .NET Framework
- .NET Core
- Mono
- Xamarin
- UWP
- Unity

.NET Standard

There are different versions of the .NET Standard, with version 2.1 being the latest and final release of the specification. The higher the version number, the more APIs are supported.

However, .NET Standard is not relevant anymore for modern .NET development.

Beginning with .NET 5 and the unification of .NET, we have a different philosophy and do not need .NET Standard anymore.

A class library targeting .NET 5 or higher can be shared across all modern .NET application types.

.NET Standard

.NET Standard version	1	1.1	1.2	1.3	1.4	1.5	1.6	2	2.1
.NET Core and .NET	1	1	1	1	1	1	1	2	3
.NET Framework	4.5	4.5	4.5.1	4.6	4.6.1	4.6.2		4.7.1	
Mono	4.6	4.6	4.6	4.6	4.6	4.6	4.6	5.4	6.4
UWP	10	10	10	10	10	10.0.16299	10.0.16299	10.0.16299	
Windows	8	8	8.1						
Windows Phone	8.1	8.1	8.1						

.NET Standard

// Defines the APIs available in .NET Standard 1.0i

interface NETStandard1_0

 { void SomeMethod();} // .NET Standard 1.1

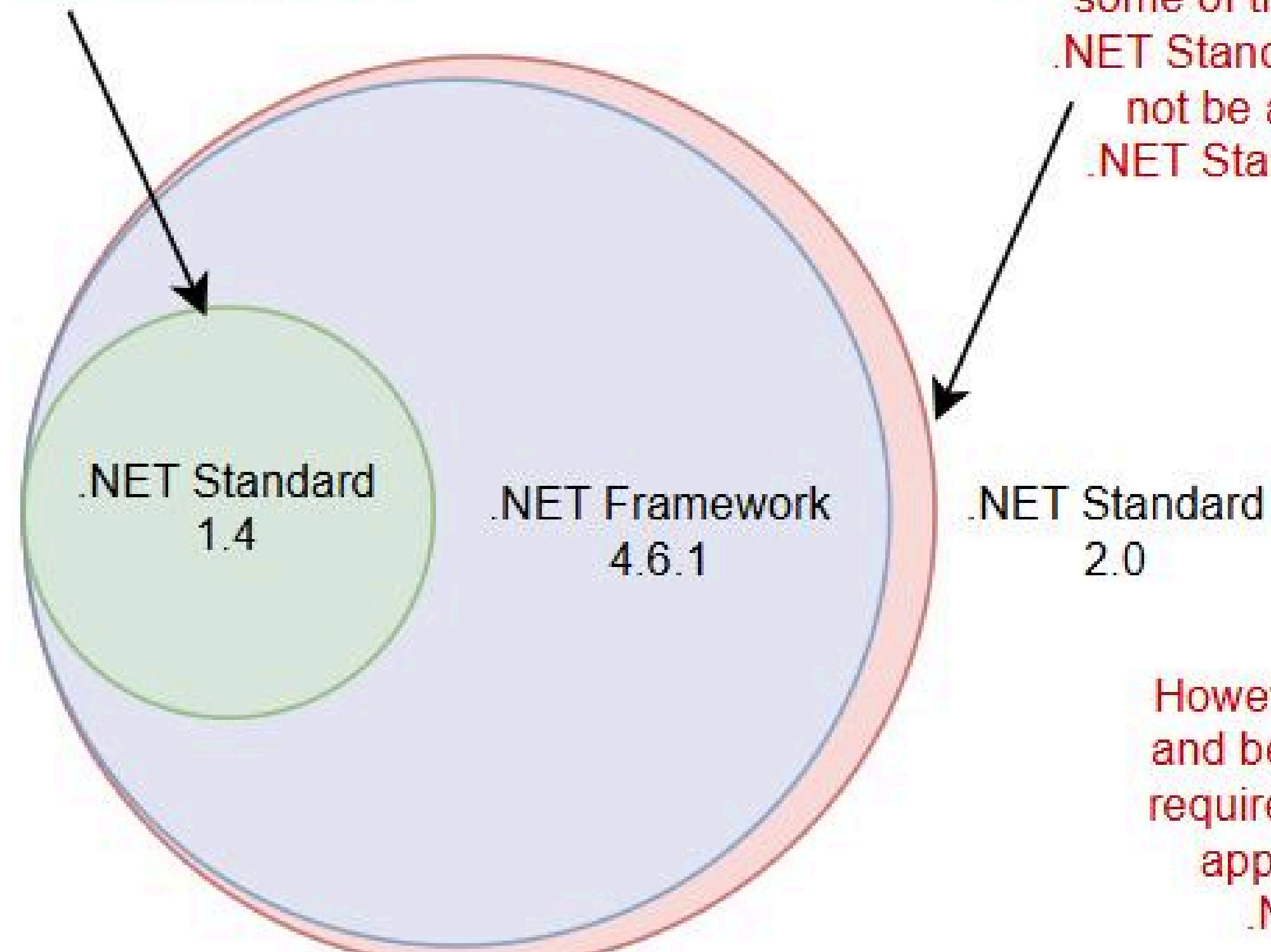
inherits all the APIs from .NET Standard 1.0.

interface NETStandard1_1 : NETStandard1_0

 { void OtherMethod();}

.NET Standard

.NET Framework 4.6.1 contains
all the required APIs so it supports
.NET Standard 1.4.



.NET Framework 4.6.1 is missing
some of the APIs required by
.NET Standard 2.0, so it should
not be able to reference
.NET Standard 2.0 libraries.

However, in order to speed adoption,
and because it contains nearly all the
required APIs, .NET Framework 4.6.1
apps and libraries can reference
.NET Standard 2.0 libraries.

Example of dot net standard

.NET STANDARD EXPLAINED

A common interface of APIs for any .NET implementation

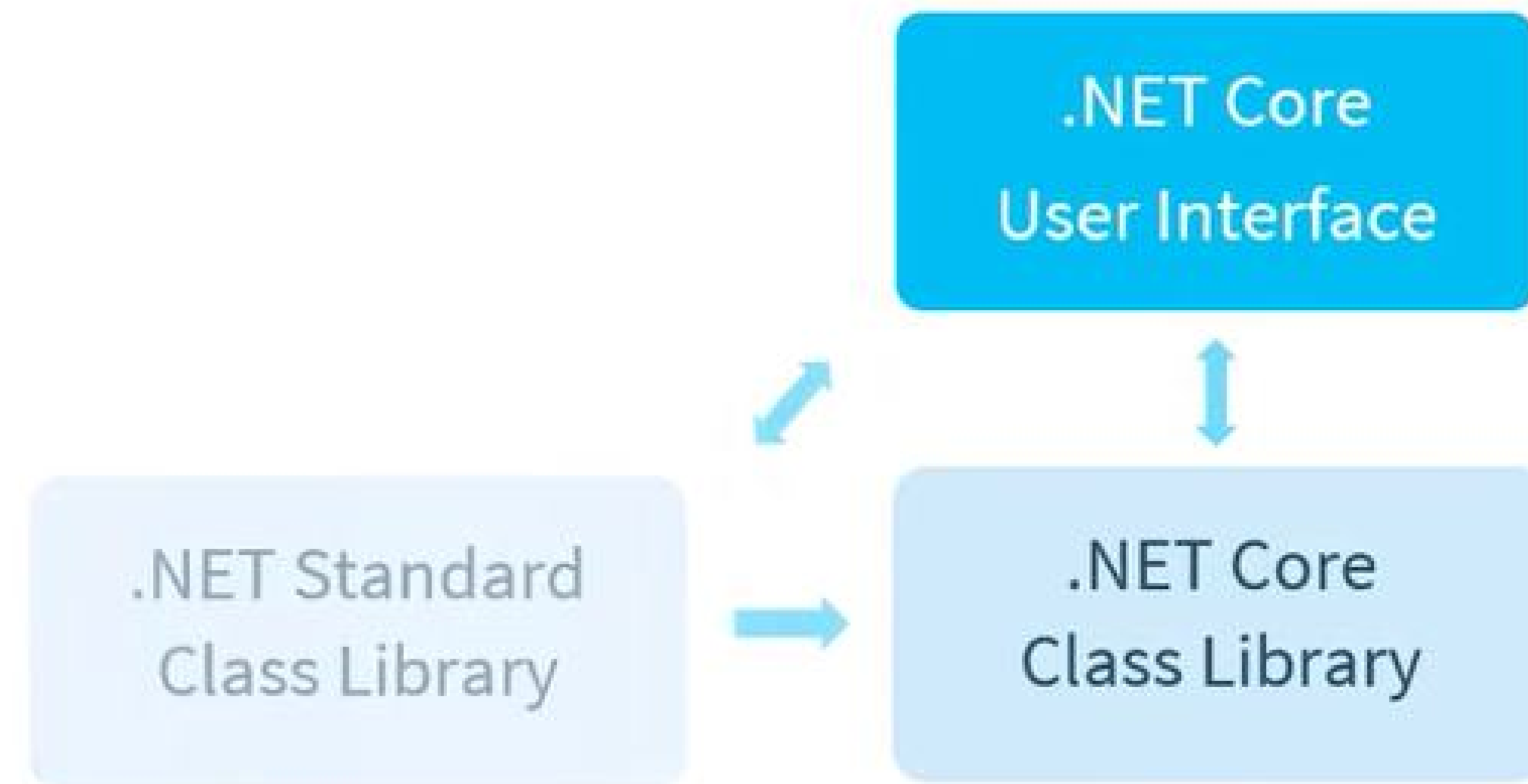


.NET Standard

A common interface of APIs for any .NET implementation



.NET Standard



Base Class Library (BCL)

System.Web

System.Windows.Form

System.Drawing

System.Data

System.Xml

System

Collections

IO

Security

Runtime

Configuration

Net

ServiceProcess

InteropServices

Diagnostics

Reflection

Text

Remoting

Globalization

Resources

Threading

Serialization